SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 3.1: Statistics I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Statistics I
Driving Question	DRIVING QUESTION: How can we use statistics to tell an honest and useful story about data from our own Kenyan community — and help people make better decisions?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Is This Data Trying to Tell Us? — Launching the Maize Price Mystery**	Learners examined a raw, unorganised list of maize prices (in KSh per 90 kg bag) collected from markets across different Kenyan counties (e.g., Eldoret, Kisumu, Nakuru, Nairobi). They attempted to answer the question: 'Is the price fair?' using only the raw list, and recorded what they could — and could not — easily determine.	Learners discovered that raw, unorganised data cannot settle a real-life disagreement fairly. They could identify the minimum and maximum price by scanning the list, but could not quickly determine how prices were distributed, which price was most common, or what a 'typical' price looked like. This created a felt need for organising tools.	The teacher should use this lesson primarily to generate productive confusion — do NOT resolve the mystery yet. Cold-call at least 3 learners to share what they noticed and what frustrated them about the raw list. Annotate the class Driving Question Board (DQB) with learner-generated questions. Key pedagogical note: the 'mystery' (why do maize prices vary so much across Kenya?) should remain open and anchor all subsequent lessons. Learners who find min/max quickly should be pushed: 'That tells us the extremes — but does it tell a farmer in Kericho what price to expect?'
Lesson 2 **Meeting the Maize Price Data: What Is Raw Data and Where Does It Come From?**	Learners examined two versions of the same maize price information: one collected directly by a fictional student journalist from Nakuru market (primary source) and one extracted from a Kenya National Bureau of Statistics (KNBS) agricultural price report (secondary source). They compared the two datasets for detail, reliability, and ease of collection.	Raw data is data that has not yet been organised or summarised. Data can come from primary sources (collected directly, e.g., surveying traders at Wakulima Market, Nairobi) or secondary sources (already compiled, e.g., KNBS reports, FAO databases). Each source type has advantages and limitations: primary data is specific but costly to collect; secondary data is convenient but may not match the exact question being investigated.	Teacher should explicitly connect source type to data trustworthiness — a core statistical literacy skill. Ask: 'If a maize trader in Eldoret and the KNBS both give you a price figure for the same month, but the numbers differ, which do you trust — and why?' Allow wait time of at least 15 seconds before cold-calling. Reinforce that understanding where data comes from is the first step in 'telling an honest story.' Update the DQB: learners should now be

			able to answer the sub-question 'Where did our maize price data come from?'
Lesson 3 **Organising the Chaos: Frequency Distribution Tables for Ungrouped Data**	Learners were given a manageable dataset of 30 ungrouped maize prices (discrete values, e.g., prices rounded to the nearest KSh 100) and physically sorted the values using tally marks on a blank table template. They counted tallies to produce frequency counts and checked that all frequencies summed to the total number of data points (n = 30).	A frequency distribution table for ungrouped data organises raw data by listing each distinct value alongside a tally and its frequency count. The value with the highest frequency is the mode. The sum of all frequencies must equal n (the total number of data values). This structure makes it immediately visible which maize prices occurred most often and which were rare.	Teacher should circulate during the tally activity and check two common errors: (1) learners miscounting tally groups of five, and (2) learners omitting values that appear zero times when constructing a complete table. Use a think-pair-share to have learners verify each other's frequency totals before class discussion. Key question for whole-class: 'A trader says the most common maize price is KSh 3,200. How does our table help us confirm or challenge that claim?' This directly reconnects to the driving question's theme of honesty in data storytelling.
Lesson 4 **Frequency Distribution Tables for Ungrouped Data — Bringing Order to the Maize Price Mystery**	Learners applied their frequency table skills to a larger ungrouped dataset (n = 50 maize prices) and used the completed table to identify the mode, interpret the distribution shape (skewed or symmetric), and make a written claim — with evidence — about whether maize prices were consistent or variable across the sampled markets.	Learners consolidated that: (i) frequency means the number of times a value occurs in a dataset; (ii) a frequency distribution table organises raw data into a clear structure that reveals patterns invisible in the raw list; (iii) the mode is the value with the highest frequency; (iv) the table is an honest summary because no data values are discarded — every observation is represented. Learners also practised writing evidence-based claims, a key competency for the driving question.	This is a consolidation lesson — resist the urge to introduce new content. Focus assessment on whether learners can justify claims using the table, not just read values off it. Suggested cold-call question: 'Without looking at the raw data again, can you tell me how many traders charged exactly KSh 3,500? Where in the table is your evidence?' Push learners to use precise language: 'The frequency table shows…' rather than vague statements. Update the DQB to reflect that learners can now answer: 'Which maize price appeared most often in our sample?'
Lesson 5 **Histograms: Making the Maize Price Distribution Visible**	Learners constructed histograms from grouped maize price data, plotting frequency density (frequency ÷ class width) on the y-axis and price class intervals on the x-axis. They compared their histogram to a bar chart of the same data and identified the key structural difference: bars in a histogram are contiguous (touching) because the data is continuous, and area — not height — represents frequency.	Learners learned that: (i) a histogram uses bars whose AREA represents frequency, so frequency density (frequency ÷ class width) must be plotted on the y-axis, especially when class widths are unequal; (ii) bars are drawn with no gaps because the underlying variable (price) is continuous; (iii) the shape of the histogram reveals the distribution of prices — whether most prices cluster in a narrow range or spread widely; (iv) a histogram is different from a bar chart, which is used for categorical (discrete/non-numeric) data.	Teacher should address the most common misconception directly: many learners plot raw frequency on the y-axis instead of frequency density. Use a counterexample with unequal class widths (e.g., one class KSh 3,000–3,500 and another KSh 3,500–4,500) to show why this produces a misleading picture. Ask: 'If we used frequency on the y-axis and the second class is twice as wide, what dishonest story would the histogram tell a maize farmer?' This connects powerfully to the driving question's honesty theme. Require learners to label axes fully (including units) before a histogram is considered complete.

Lesson 6 **Grouping the Maize Price Data: Grouped Frequency Tables and the Modal Class**	Learners grouped the full maize price dataset into class intervals of equal width (e.g., KSh 3,000–3,199, 3,200–3,399, etc.), constructed a grouped frequency distribution table, and identified the modal class. They then compared the grouped table to their earlier ungrouped table and discussed what information was gained and what was lost.	Learners learned that: (a) grouping raw data into class intervals makes the distribution pattern more visible, especially for large datasets, but individual values are lost — you can no longer say exactly how many traders charged KSh 3,250; (b) the modal class is the class interval with the highest frequency, not a single value; (c) class intervals must be mutually exclusive (no overlap) and collectively exhaustive (every value fits somewhere); (d) choosing an appropriate class width is a judgement call — too few classes hide pattern, too many classes restore the chaos of raw data.	Teacher should emphasise the trade-off between precision and clarity — this is a genuine statistical judgement, not a right/wrong procedure. Pose the decision-making scenario: 'You are advising the Nakuru County government on setting a fair maize price ceiling. Would you rather have the ungrouped table or the grouped table? Why?' Cold-call 3 learners with different answers and use their reasoning to build a class consensus. Reinforce that the modal class gives a range, not a single 'most common price,' which affects how confidently a claim can be made.
Lesson 7 **Measures of Central Tendency for Grouped Data — Cracking the Maize Price Mystery at Scale**	Learners used the grouped frequency distribution table of maize prices to estimate the mean using the assumed-mean (deviation/working mean) method, identify the modal class and use its midpoint as the estimated mode, and locate the median class using cumulative frequencies. They then interpreted all three measures together and discussed which best represented the 'typical' maize price a Kenyan farmer or consumer would encounter.	Learners learned how to: (i) estimate the mean of grouped data using the assumed-mean/deviation method (mean = $A + \Sigma fd/\Sigma f$, where d = midpoint − assumed mean); (ii) identify the modal class and report its midpoint as the estimated mode; (iii) locate the median class as the class containing the $(n/2)$th value using a cumulative frequency column. Learners also learned that the three measures can give different values for the same dataset, and that choosing which measure to report is a statistical and ethical decision — directly linking to the driving question.	Teacher should stress that the assumed-mean method is computationally equivalent to the direct method but reduces arithmetic errors with large price values (e.g., working with deviations from KSh 3,400 rather than the raw prices). A common error is using class boundaries instead of midpoints — intercept this early by requiring learners to write out the midpoint column before any calculation begins. The synthesis question for this lesson: 'A news headline reads: The average maize price in Kenya is KSh X. Based on our analysis, is that headline honest and useful? What else should it include?' This is a direct answer attempt to the driving question and should be recorded on the DQB as a cumulative model update.
Lesson 8 **Telling an Honest Story — Presenting Our Kenyan Maize Price Findings**	Learners compiled all their statistical tools — raw data source evaluation, frequency distribution tables (ungrouped and grouped), histograms, and measures of central tendency — into a short, structured data report on the maize price dataset. They presented their findings to the class in the form of a statistical brief addressed to a fictional Kenyan county agricultural officer, justifying their choice of summary measure and acknowledging the limitations of their analysis.	Learners synthesised the full sub-strand by demonstrating that: (i) statistics is a process — from raw data collection through organisation, visualisation, and summarisation to interpretation; (ii) every statistical choice (grouping, measure of central tendency, graph type) involves a judgement that affects the story the data tells; (iii) an honest and useful statistical report names its data source, uses appropriate measures, displays data clearly, and acknowledges what the data cannot show. These competencies directly answer the driving question.	Teacher should frame this lesson as a performance task, not a test. Assess learners on their ability to justify choices, not just compute correctly — e.g., 'Why did you report the median rather than the mean for this dataset?' is more revealing than checking arithmetic. Provide sentence starters for learners who struggle with written justification: 'I chose to use [measure] because…'; 'This histogram shows that…'; 'One limitation of our data is…'. Close the lesson by returning to the original DQB from Lesson 1 and having the class vote on which sub-questions they can

			now confidently answer — this gives learners a visible sense of cumulative learning across the full sub-strand.